Validity Analysis: LAILA

Target context: Arab-Country G10–12 Arabic Teachers: AI Essay Grading Assistance

Overall risk: **HIGH**

Deployment Context

Use case: Assist high school (G10-12) Arabic teachers in Arab countries (teaching students who are native in Arabic) in grading Arabic essays (dataset was collected in Qatar but the writing is in modern standard Arabic which is standard across all Arab countries, although the standards for grading may differ from country to country).

Target population: Arabic teachers in Arab countries teaching high school students (G10-12)

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology ⚠	2	Significant gaps	high
Input Content ⚠	2	Significant gaps	high
Input Form ✓	4	Minor gaps	high
Output Ontology ⚠	2	Significant gaps	high
Output Content ✓	3	Moderate gaps	medium
Output Form ⚠	1	Serious concern	high
Average	**2.3**		

⚠ = highest concern ✓ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

LAILA is a methodologically rigorous, well-documented Arabic AES benchmark that establishes the largest publicly available trait-annotated Arabic essay dataset to date [Q1, Q2]. For the target deployment — pan-Arab G10–12 Arabic teachers using AI for essay grading assistance — its validity is mixed. Input Form is well aligned (text-only MSA, matching modality). Output Content has strong annotation procedures but unresolved questions about annotator nationality diversity. The remaining four dimensions show significant deployment mismatches that map directly onto the HIGH-priority gaps flagged in elicitation: Input Ontology (Egypt's holistic grading [WEB-5] and missing genres [Q89]), Input Content (Qatari-only prompts [Q88] with no cross-national degradation testing), Output Ontology (no confidence or rationale categories), and Output Form (discrete labels only, with the benchmark explicitly prohibiting high-stakes operational use without independent validation [Q104, Q105]). The benchmark is best understood as a strong research resource that requires substantial supplementation — both empirical (cross-national validation) and methodological (rationale and uncertainty layers) — before responsible teacher-facing deployment.

ID	Evidence
Q1	“LAILA, the largest publicly available Arabic AES dataset to date, comprising 7,859 essays annotated with holistic and trait-specific scores on seven dimensions: relevance, organization, vocabulary, style, development, mechanics, and grammar.” (p.1)
Q2	“The dataset comprises 7,859 essays across 8 distinct prompts making it the most extensive publicly available Arabic resource of its kind.” (p.1)
Q88	“First, LAILA was collected from students in a single country, Qatar, and specific grade levels, grades 10 to 12, which may limit its generalizability to other Arabic-speaking populations due to diverse educational systems. However, this concern is partially mitigated by Qatar's large population of Arab expatriates, resulting in student participants representing a wide variety of Arabic dialects and backgrounds.” (p.9)
Q89	“Second, LAILA includes only explanatory and persuasive writing prompts, limiting genre diversity and potentially affecting model robustness across other styles, such as narrative or descriptive writing.” (p.10)
Q104	“We explicitly prohibit its use for high-stakes educational decisions affecting individual students.” (p.10)
Q105	“Users must commit to: (1) preventing re-identification attempts, (2) avoiding demographic profiling or inference, and (3) refraining from deployment in operational assessment contexts without independent validation.” (p.10)
WEB-5	Egypt's holistic grading culture suggests Egyptian teachers may apply different evaluative standards than Qatar-based annotators (https://grokipedia.com/page/academic_grading_in_egypt)

Practical Guidance

What This Benchmark Measures

LAILA measures how well models can predict CAST-rubric trait scores and a derived holistic score on explanatory and persuasive MSA essays produced by G10–12 Qatari students under timed digital conditions [Q14, Q34, Q38]. Its strongest dimensions for the target context are Input Form (text-only MSA matches deployment) and Output Content (rigorous annotation with substantial IAA [Q44]), making it useful for benchmarking baseline scoring agreement between models and qualified Arabic-educator judgment.

Construct Depth

The benchmark probes the construct of Arabic essay quality with substantial depth on the analytic-rubric dimension — fine-grained seven-trait scoring, double annotation, rigorous adjudication, and extensive baseline experiments across feature-based, encoder-based, and LLM models in both prompt-specific and cross-prompt setups [Q48, Q83]. However, depth is concentrated within a Qatari-curriculum, two-genre, discrete-label paradigm. It does not probe construct depth on dimensions central to the deployment: cross-national rubric portability (Egypt's holistic grading, Levantine and Maghrebi conventions), narrative/descriptive genre coverage, uncertainty calibration, or rationale quality.

What Else You Need

For valid teacher-facing deployment beyond Qatar, supplementation is required across four dimensions: (IO) cross-national rubric mapping plus a holistic-only evaluation track for Egyptian-style use; (IC) cross-national prompt validation testing model scoring stability on Egyptian, Levantine, and Maghrebi locally-grounded prompts; (OO+OF) integration of an Arabic-language adaptation of rationale-generation frameworks such as RaDME or RMTS [WEB-13, WEB-12] plus calibrated uncertainty estimates evaluated against teacher disagreement; (OC) a cross-country annotator validation study — re-annotating a stratified sample of LAILA essays with Egyptian, Saudi, Jordanian, Lebanese, and Moroccan Arabic-language educators to quantify cross-country labeling drift on Style, Development, and Organization. The benchmark's own ethics requirements [Q105, Q106] effectively mandate this supplementation before operational deployment.

ID	Evidence
Q14	“We establish baseline AES results for Arabic under two evaluation setups: prompt-specific and cross-prompt.” (p.2)

Q34	“The rubric covers 7 writing traits [D4]: Relevance (REL), Organization (ORG), Vocabulary (VOC), Style (STY), Development (DEV), Mechanics (MEC), and Grammar (GRA).” (p.5)
Q38	“LAILA comprises 7,859 essays collected across 8 distinct prompts: 3 explanatory and 5 persuasive (3,446 and 4,413 essays, respectively).” (p.6)
Q44	“We compute IAA using Quadratic Weighted Kappa (QWK) (Williamson et al., 2012) to assess agreement between the two main annotators (A1 and A2), prior to adjudication. Table 4 reports QWK per trait and prompt, with agreement strength following Landis and Koch (1977). Overall, IAA was substantial across prompts, with an average QWK ranging from 0.66 (P5) to 0.75 (P1), showing strong consistency.” (p.6)
Q48	“The models fall into three categories: feature-based (FB), encoder-based, and large language models (LLMs). LLMs were evaluated in zero-shot and few-shot settings, and all models followed a multitask setup predicting holistic and trait scores jointly.” (p.7)
Q83	“We also benchmarked LAILA using SOTA Arabic and English AES models in both prompt-specific and cross-prompt settings, showing strong baselines for future research.” (p.9)
Q105	“Users must commit to: (1) preventing re-identification attempts, (2) avoiding demographic profiling or inference, and (3) refraining from deployment in operational assessment contexts without independent validation.” (p.10)
Q106	“We recommend that any system developed using LAILA undergo fairness auditing and stakeholder consultation before real-world application.” (p.10)
WEB-12	BERT-based Arabic AES system documented but without rationale generation (https://www.researchgate.net/publication/382302599_Automated_essay_scoring_in_Arabic_a_dataset_and_analysis_of_a_BERT-based_system)
WEB-13	RaDME generates trait scores with co-produced rationales using knowledge distillation — English only, no Arabic adaptation documented (https://arxiv.org/html/2502.20748v1)

Input Ontology (IO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: *LAILA* | ***Context:*** *Arab-Country G10–12 Arabic Teachers: AI Essay Grading Assistance*

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

LAILA's task taxonomy is explicitly limited to two genres (explanatory and persuasive) [Q89] and operationalizes essay scoring through the QUTC CAST seven-trait rubric [Q33, Q34], a Qatari-curriculum artifact. For a pan-Arab teacher-facing deployment, this presents two HIGH-priority IO problems flagged in the elicitation: (1) Egypt's secondary Arabic assessment is holistic and rubric-free with no documented multi-trait analogue [WEB-5]; the seven-trait structure does not map onto Egyptian classroom practice. (2) Narrative and descriptive genres, which are present in many Arab national curricula, are absent from the benchmark and are explicitly noted as a limitation by the authors [Q89]. The benchmark also frames AES purely as score prediction (regression / sequential generation of trait labels) [Q147, Q181] and does not include any task category for confidence estimation or rationale generation, both of which the deployment requires. The two evaluation setups (prompt-specific and cross-prompt) [Q14] are methodologically appropriate but do not address cross-national generalization. No public rubric documentation was found for Saudi Arabia, Jordan, Lebanon, or Morocco that would allow the seven-trait mapping to be confirmed [WEB-7, WEB-10]. The strengths are real (multi-trait coverage, two evaluation setups), but the omissions are central to deployment validity.

ID	Evidence
Q14	“We establish baseline AES results for Arabic under two evaluation setups: prompt-specific and cross-prompt.” (p.2)
Q33	“All essays in LAILA dataset were scored using the Core Academic Skills Test (CAST) rubric [D7], developed by the Qatar University Testing Center (QUTC).” (p.5)
Q34	“The rubric covers 7 writing traits [D4]: Relevance (REL), Organization (ORG), Vocabulary (VOC), Style (STY), Development (DEV), Mechanics (MEC), and Grammar (GRA).” (p.5)
Q89	“Second, LAILA includes only explanatory and persuasive writing prompts, limiting genre diversity and potentially affecting model robustness across other styles, such as narrative or descriptive writing.” (p.10)
Q147	“The AES task is formulated as a text generation problem, where the model predicts the trait scores sequentially.” (p.16)
Q181	“In the zero-shot setting, the LLM is tasked with generating trait scores in JSON format, given the prompt text, essay, and trait rubrics.” (p.18)
WEB-5	Egypt's holistic grading culture suggests Egyptian teachers may apply different evaluative standards than Qatar-based annotators (https://grokipedia.com/page/academic_grading_in_egypt)
WEB-7	Saudi Qiyas focuses on higher-education entry testing rather than secondary multi-trait classroom rubrics (https://etec.gov.sa/en/centers/qiyas)
WEB-10	Jordan's Tawjihi Arabic exam emphasizes literary comprehension and grammar over trait-scored free composition (https://en.wikipedia.org/wiki/Tawjihi)

Strengths

- Provides fine-grained seven-trait coverage (REL, ORG, VOC, STY, DEV, MEC, GRA) supporting analytic rather than only holistic scoring [Q34, Q115]
- Includes both prompt-specific and cross-prompt evaluation setups, with cross-prompt being the more deployment-realistic case [Q14, Q72]
- Evaluates a diverse model taxonomy (feature-based, encoder-based, Arabic-centric LLMs in zero/few-shot) [Q48, Q51], giving teachers a meaningful baseline landscape

ID	Evidence
Q14	"We establish baseline AES results for Arabic under two evaluation setups: prompt-specific and cross-prompt." (p.2)
Q34	"The rubric covers 7 writing traits [D4]: Relevance (REL), Organization (ORG), Vocabulary (VOC), Style (STY), Development (DEV), Mechanics (MEC), and Grammar (GRA)." (p.5)
Q48	"The models fall into three categories: feature-based (FB), encoder-based, and large language models (LLMs). LLMs were evaluated in zero-shot and few-shot settings, and all models followed a multitask setup predicting holistic and trait scores jointly." (p.7)
Q51	"For LLMs, we evaluated three Arabic-centric models: ALLaM (Bari et al., 2025), Command-R7B-Arabic (R7B) (Alnumay et al., 2025), and Fanar (Team et al., 2025), under zero-shot and few-shot (5-shot) settings." (p.7)
Q72	"We note that the applicability of the prompt-specific setup is relatively limited due to reliance on target prompt essays that are labeled, which is often unavailable in practice." (p.9)
Q115	"This rubric guided scoring across 7 writing traits: relevance (REL), organization (ORG), vocabulary (VOC), style (STY), development (DEV), mechanics (MEC), and grammar (GRA)." (p.12)

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required deployment categories include: (a) trait-based analytic scoring for Gulf-style rubric users, (b) holistic-only scoring suited to Egyptian practice, (c) narrative and descriptive genre scoring, and (d) confidence/rationale outputs. LAILA covers (a) explicitly [Q34] but does not cover (b), (c), or (d).

IO-2: Yes. The taxonomy omits narrative and descriptive genres [Q89], omits a holistic-only scoring mode aligned with Egyptian practice [WEB-5], and omits confidence/rationale categories that the deployment requires [elicitation Q3]. No mapping to Tawjihi (Jordan) or Qiyas (Saudi Arabia) frameworks is documented [WEB-7, WEB-10].

IO-3: No clearly irrelevant categories were identified — all seven CAST traits are plausibly meaningful for MSA writing, even if not all are weighted equivalently across national curricula. The pan-Arab MSA standards assumption from elicitation supports surface relevance of the trait set.

IO-4: Documented gaps that harm content validity: (1) genre ontology restricted to explanatory/persuasive [Q89]; (2) no holistic-only or rubric-free evaluation mode for Egyptian-style use [WEB-5]; (3) no task category for uncertainty or rationale generation [Q181, Q182]; (4) no cross-national rubric mapping between CAST traits and Saudi/Jordanian/Lebanese/Moroccan frameworks [WEB-7, WEB-10].

ID	Evidence
Q3	"Under Institutional Review Board (IRB) approval, we visited 24 high schools in Qatar and collected essays directly from 4,372 students in authentic classroom settings." (p.1)
Q34	"The rubric covers 7 writing traits [D4]: Relevance (REL), Organization (ORG), Vocabulary (VOC), Style (STY), Development (DEV), Mechanics (MEC), and Grammar (GRA)." (p.5)
Q89	"Second, LAILA includes only explanatory and persuasive writing prompts, limiting genre diversity and potentially affecting model robustness across other styles, such as narrative or descriptive writing." (p.10)
Q181	"In the zero-shot setting, the LLM is tasked with generating trait scores in JSON format, given the prompt text, essay, and trait rubrics." (p.18)
Q182	"The holistic score was computed as the sum of the individual trait scores." (p.18)

WEB-5	Egypt's holistic grading culture suggests Egyptian teachers may apply different evaluative standards than Qatar-based annotators (https://grokipedia.com/page/academic_grading_in_egypt)
WEB-7	Saudi Qiyas focuses on higher-education entry testing rather than secondary multi-trait classroom rubrics (https://etec.gov.sa/en/centers/qiyas)
WEB-10	Jordan's Tawjihi Arabic exam emphasizes literary comprehension and grammar over trait-scored free composition (https://en.wikipedia.org/wiki/Tawjihi)

Information Gaps

- Genre distribution (narrative/descriptive/argumentative/explanatory) in G10–12 Arabic curricula across Egypt, Saudi Arabia, Jordan, Lebanon, Morocco
- Whether national curricula in target countries use any analytic trait dimensions that would map onto CAST

Requires Expert Verification

- Mapping of CAST traits to Lebanese CRDP and Moroccan national Arabic writing criteria
- Whether Saudi MOE classroom Arabic assessment uses any internal trait-based rubric beyond Qiyas

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Remediation

Gap 1: Seven-trait CAST taxonomy does not map onto Egyptian holistic grading practice and lacks narrative/descriptive genre coverage

Recommendation: Add a holistic-only evaluation track suitable for Egyptian-style assessment, and either commission a narrative/descriptive prompt extension or document explicitly that the system is restricted to explanatory/persuasive genres in deployment guidelines.

Gap 2: Cross-prompt generalization performance is substantially weaker than prompt-specific performance [Q71], yet cross-prompt is the realistic deployment setting [Q72]

Recommendation: Anchor deployment evaluation primarily on cross-prompt results rather than prompt-specific results, and communicate this performance ceiling clearly to teacher users so they understand the system's reliability limits on novel prompts.

ID	Evidence
Q71	“However, its cross-prompt performance drops significantly (average QWK: 0.549), underscoring its ability to capture prompt-dependent patterns rather than generalizable features.” (p.9)
Q72	“We note that the applicability of the prompt-specific setup is relatively limited due to reliance on target prompt essays that are labeled, which is often unavailable in practice.” (p.9)

Input Content (IC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: *LAILA* | ***Context:*** *Arab-Country G10–12 Arabic Teachers: AI Essay Grading Assistance*

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

All 7,859 essays were collected from 4,372 students at 24 schools in Qatar [Q3, Q27], in response to 8 prompts designed for the Qatari educational context. The authors themselves flag that this 'may limit its generalizability to other Arabic-speaking populations due to diverse educational systems' [Q88], offering only partial mitigation through Qatar's expatriate population. Prompts were designed for cultural relevance to the Arabic-speaking context broadly [Q22, Q99], but no prompts grounded in Egyptian history, Levantine literary traditions, or North African social topics are documented. The elicitation flags IC as HIGH priority because models trained on LAILA may misjudge Relevance and Development for locally grounded non-Qatari content (Q4 elicitation), and web searches found no published study testing LAILA-trained models on non-Qatari prompt content (gap_id 2). Comparison datasets cited (ASAP, PERSUADE, MERLIN, TOEFL11) [Q17, Q112, Q113] situate LAILA internationally but not within a cross-Arab-country framework. The content is authentically classroom-produced [Q21] and gender-balanced [Q23, Q100], which are real strengths for L2/L1 MSA-writer representation, but the geographic single-country scope is the dominant validity concern.

ID	Evidence
Q3	"Under Institutional Review Board (IRB) approval, we visited 24 high schools in Qatar and collected essays directly from 4,372 students in authentic classroom settings." (p.1)
Q4	"Table 1: Brief description of the traits in LAILA." (p.1)
Q17	"Research on AES has advanced considerably in English, supported by large-scale, publicly available datasets such as ASAP/ASAP++ (Mathias and Bhattacharyya, 2018), TOEFL11 (Blanchard et al., 2013), ELLIPSE (Crossley et al., 2023a), and PERSUADE (Crossley et al., 2023b)." (p.2)
Q21	"[D1] Ensuring Data Integrity: We designed LAILA to capture authentic, classroom-produced writing created without external assistance. This design choice preserves genuine linguistic variation, learner errors, and authentic complexities essential for real-world AES applications." (p.3)
Q22	"[D2] Maximizing Diversity: To reflect the breadth of Arabic writing proficiency, LAILA was planned to collect essays from multiple educational institutions and student populations with diverse academic and demographic backgrounds. Prompts were intentionally varied across genres, ensuring cultural relevance and age appropriateness, thereby promoting model generalization and reducing bias." (p.3)
Q23	"[D3] Achieving Gender Balance: We intentionally aimed to balance the number of essays from male and female students to promote fairness, reduce gender-based bias, support inclusive evaluation, and enhance model generalization." (p.3)
Q27	"Data collection spanned one academic year through 27 school visits (24 initial, three follow-ups), by nine team members, to meet the target student count." (p.5)
Q88	"First, LAILA was collected from students in a single country, Qatar, and specific grade levels, grades 10 to 12, which may limit its generalizability to other Arabic-speaking populations due to diverse educational systems. However, this concern is partially mitigated by Qatar's large population of Arab expatriates, resulting in student participants representing a wide variety of Arabic dialects and backgrounds." (p.9)
Q99	"Essay prompts were carefully designed to be developmentally appropriate for the target age groups, culturally relevant to the Arabic-speaking context, and free of sensitive topics or potentially harmful content." (p.10)
Q100	"The dataset includes balanced gender representation and covers grades 10-12." (p.10)
Q112	"The MERLIN corpus: Learner language and the CEFR." (p.11)
Q113	"TOEFL11: A corpus of non-native English." (p.11)

Strengths

- Authentic classroom-produced writing under realistic timed conditions [Q21, Q25]
- Intentional gender balance and diversity-by-design principles [Q23, Q100]
- Substantial scale (7,859 essays, 4,372 students, 24 schools, 8 prompts) [Q2, Q27, Q38]
- Qatar's Arab expatriate population provides some dialectal and cultural diversity in student-writer backgrounds [Q88]

ID	Evidence
Q2	"The dataset comprises 7,859 essays across 8 distinct prompts making it the most extensive publicly available Arabic resource of its kind." (p.1)
Q21	"[D1] Ensuring Data Integrity: We designed LAILA to capture authentic, classroom-produced writing created without external assistance. This design choice preserves genuine linguistic variation, learner errors, and authentic complexities essential for real-world AES applications." (p.3)
Q23	"[D3] Achieving Gender Balance: We intentionally aimed to balance the number of essays from male and female students to promote fairness, reduce gender-based bias, support inclusive evaluation, and enhance model generalization." (p.3)
Q25	"[D5] Ensuring Authentic Writing Conditions: To emulate real academic or high-stakes testing scenarios, LAILA was designed to collect essays written under timed conditions using a digital submission platform. This setup ensures consis" (p.3)
Q27	"Data collection spanned one academic year through 27 school visits (24 initial, three follow-ups), by nine team members, to meet the target student count." (p.5)
Q38	"LAILA comprises 7,859 essays collected across 8 distinct prompts: 3 explanatory and 5 persuasive (3,446 and 4,413 essays, respectively)." (p.6)
Q88	"First, LAILA was collected from students in a single country, Qatar, and specific grade levels, grades 10 to 12, which may limit its generalizability to other Arabic-speaking populations due to diverse educational systems. However, this concern is partially mitigated by Qatar's large population of Arab expatriates, resulting in student participants representing a wide variety of Arabic dialects and backgrounds." (p.9)
Q100	"The dataset includes balanced gender representation and covers grades 10-12." (p.10)

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes. Prompts are grounded in the Qatari educational context [Q88] and would require region-specific cultural framing for relevance/development scoring of essays from other Arab countries [elicitation Q4]. No Egyptian-history, Levantine-literary, or North-African-social prompts are documented.

IC-2: Partially. Prompts were designed to be 'culturally relevant to the Arabic-speaking context' and free of sensitive topics [Q22, Q99], which provides broad pan-Arab cultural compatibility. However, alignment is to a Qatari-designed conception of pan-Arab cultural relevance, not to nationally specific curricula.

IC-3: No Western-specific knowledge is required; the content is Arabic-language and Arab-context throughout [Q22]. The risk is intra-region (Qatar-centric within Arab context) rather than Western-bias.

IC-4: INSUFFICIENT DOCUMENTATION — the paper does not document any regional annotator panel from non-Qatari Arab countries reviewing prompts for cross-national cultural representativeness; would need an Egyptian/Levantine/Maghrebi expert review of the eight prompts.

IC-5: Documented content issues: (1) single-country source [Q88]; (2) eight prompts may not span the cultural and topical breadth of seven-plus deployment countries; (3) no published cross-national degradation test exists [gap_id 2]. The HIGH-priority IC mismatch flagged in elicitation is corroborated by both the authors' own limitations statement and the absence of compensating evidence.

ID	Evidence
Q4	"Table 1: Brief description of the traits in LAILA." (p.1)
Q22	"[D2] Maximizing Diversity: To reflect the breadth of Arabic writing proficiency, LAILA was planned to collect essays from multiple educational institutions and student populations with diverse academic and demographic backgrounds. Prompts were intentionally varied across genres, ensuring cultural relevance and age appropriateness, thereby promoting model generalization and reducing bias." (p.3)
Q88	"First, LAILA was collected from students in a single country, Qatar, and specific grade levels, grades 10 to 12, which may limit its generalizability to other Arabic-speaking populations due to diverse educational systems. However, this concern is partially mitigated by Qatar's large population of Arab expatriates, resulting in student participants representing a wide variety of Arabic dialects and backgrounds." (p.9)
Q99	"Essay prompts were carefully designed to be developmentally appropriate for the target age groups, culturally relevant to the Arabic-speaking context, and free of sensitive topics or potentially harmful content." (p.10)
WEB-6	AR-AES dataset uses undergraduate essays from diverse courses, not Arab-secondary curriculum content (https://arxiv.org/abs/2407.11212)
WEB-26	TAQEEM 2025 introduced 1,265-essay Arabic AES dataset with same seven traits; non-Qatar origin not explicitly documented in available abstracts (https://aclanthology.org/2025.arabicnlp-sharedtasks.134/)

Information Gaps

- Empirical performance of LAILA-trained models on non-Qatari prompts (gap_id 2: no study found)
- Whether TAQEEM 2025's dataset extends LAILA's geographic coverage [WEB-26]
- Dialectal-interference patterns in MSA produced by Moroccan, Egyptian, Lebanese students (gap_id 6)

Requires Expert Verification

- Cultural equivalence of the 8 LAILA prompts when assigned to Egyptian, Lebanese, or Moroccan G10–12 classrooms
- Whether Qatari prompts would be assignable as-is in non-Qatari curricula

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Remediation

Gap 1: All prompts originate in a single country (Qatar); cross-national prompt-content generalization is untested [Q88, gap_id 2]

Recommendation: Run a cross-national content validation study by collecting a small held-out set of essays responding to nationally-grounded prompts (Egyptian history, Levantine literature, Maghrebi social topics) and measure scoring stability of LAILA-trained models versus local teacher ground truth.

ID	Evidence
Q88	"First, LAILA was collected from students in a single country, Qatar, and specific grade levels, grades 10 to 12, which may limit its generalizability to other Arabic-speaking populations due to diverse educational systems. However, this concern is partially mitigated by Qatar's large population of Arab expatriates, resulting in student participants representing a wide variety of Arabic dialects and backgrounds." (p.9)

Input Form (IF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: *LAILA* | ***Context:*** *Arab-Country G10–12 Arabic Teachers: AI Essay Grading Assistance*

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark's input form is text-only Modern Standard Arabic submitted digitally via Microsoft Forms with a 65-minute time limit [Q26], producing essays averaging 171 words with most between 90 and 210 words [Q39, Q134, Q135]. PII was systematically removed and pre-generated anonymous IDs assigned [Q97]. This matches the deployment scenario directly: the elicitation marks IF as LOWER priority because deployment is text-only MSA. No script, modality, or capture-device mismatch exists. The main residual concerns are: (1) the short essay length (171-word average) is acknowledged as limiting assessment of long-form writing [Q92], and many national high-school assessments may produce longer compositions; (2) the digital-submission collection mode may not match handwritten or paper-based classroom workflows still common in some target countries — Saudi Arabia has Madrasati [WEB-17] and UAE has high digital adoption [WEB-1], but Egypt, Jordan, Lebanon, Morocco have no comparable national K-12 LMS documented in searches. The form is broadly compatible but biased toward shorter essays and digital-native workflows.

ID	Evidence
Q26	“taining one persuasive and one explanatory prompt, along with a unique, pre-generated ID to maintain students' anonymity while enabling metadata tracking [D6]. We also configured a Microsoft Form for digital essay submission, set with a 65-minute time limit to enforce a time-restricted setup [D5].” (p.5)
Q39	“Table 3 shows that the number of essays per prompt ranges from 500 to 1,181. Essay lengths demonstrate consistency across prompts, with an average of 171 words and a maximum of 706 words.” (p.6)
Q92	“Fifth, with an average essay length of only 171 words, the dataset lacks extended writing samples, restricting the assessment of models on long-form academic tasks requiring more complex, extended compositions.” (p.10)
Q97	“To protect participant privacy, all submissions were assigned pre-generated anonymous identifiers, and personally identifiable information was systematically removed during data cleaning.” (p.10)
Q134	“The data exhibits a unimodal pattern centered around an average essay length of 171 words, with the majority of essays concentrated between 90 and 210 words.” (p.15)
Q135	“A pronounced peak occurs in the 150 to 180 word range, where the count reaches its highest at 1,065 essays.” (p.15)
WEB-1	UAE teachers ~75% AI tool adoption per OECD TALIS 2024 (https://www.agbi.com/education/2025/10/uae-teachers-lead-world-in-classroom-ai-adoption/)
WEB-17	Saudi Arabia's Madrasati LMS deployed to all public K-12 schools, 92% student attendance (https://www.rowad-alkhaleej.edu.sa/en/online-learning-in-saudi-arabia/)

Strengths

- Text-only MSA matches deployment modality and Arabic script requirements [Q26]
- Authentic timed writing conditions emulate high-stakes testing [Q25]
- Systematic anonymization and PII removal supports privacy compliance [Q97]
- Comparison table situates LAILA against other AES datasets on length and annotation depth [Q12]

ID	Evidence
Q12	“Table 2: Comparison of LAILA with existing essay datasets. "Len" is average essay length in words; "HOL" indicates holistic scoring; "Eur" covers German, Italian, and Czech; "L1/L2" denotes native or second-language learners; "Public" here means freely available.” (p.2)

Q25	"[D5] Ensuring Authentic Writing Conditions: To emulate real academic or high-stakes testing scenarios, LAILA was designed to collect essays written under timed conditions using a digital submission platform. This setup ensures consis" (p.3)
Q26	"taining one persuasive and one explanatory prompt, along with a unique, pre-generated ID to maintain students' anonymity while enabling metadata tracking [D6]. We also configured a Microsoft Form for digital essay submission, set with a 65-minute time limit to enforce a time-restricted setup [D5]." (p.5)
Q97	"To protect participant privacy, all submissions were assigned pre-generated anonymous identifiers, and personally identifiable information was systematically removed during data cleaning." (p.10)

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: No script or signal-distribution mismatch — both source and target are MSA text in Arabic script [Q26]. Length distribution skews short (171-word mean, peak 150–180 [Q134, Q135]); deployment essays may be longer in some curricula, which is a partial distribution mismatch.

IF-2: Digital-submission infrastructure exists in Gulf states (Madrasati in Saudi Arabia, high UAE adoption) [WEB-17, WEB-1] but is less documented for Egypt, Jordan, Lebanon, Morocco; teachers in those countries may still receive handwritten essays requiring transcription before AI scoring.

IF-3: The CAST rubric is provided in Arabic for annotation [Q114] with English translation [Q116]; AraBERT encodes essay+prompt jointly [Q157] and AraT5 receives instruction+essay [Q148]; LLM context limit of 4,096 tokens with iterative example removal [Q184, Q185] is a relevant deployment-time form constraint.

IF-4: Documented form considerations: (1) average essay length is short (171 words) and may not represent longer compositions found in some curricula [Q92]; (2) no audio/handwriting modality despite handwritten classroom workflows being plausible in non-Gulf countries; (3) the digital Microsoft Forms collection [Q26] may not match all deployment workflows.

ID	Evidence
Q26	"taining one persuasive and one explanatory prompt, along with a unique, pre-generated ID to maintain students' anonymity while enabling metadata tracking [D6]. We also configured a Microsoft Form for digital essay submission, set with a 65-minute time limit to enforce a time-restricted setup [D5]." (p.5)
Q92	"Fifth, with an average essay length of only 171 words, the dataset lacks extended writing samples, restricting the assessment of models on long-form academic tasks requiring more complex, extended compositions." (p.10)
Q114	"For annotating LAILA, we adopted the rubric from the Core Academic Skills Test (CAST), which is provided in Arabic, developed by the Qatar University Testing Center (QUTC)." (p.12)
Q116	"An English-translated version of the CAST rubric is presented in Table 6." (p.12)
Q134	"The data exhibits a unimodal pattern centered around an average essay length of 171 words, with the majority of essays concentrated between 90 and 210 words." (p.15)
Q135	"A pronounced peak occurs in the 150 to 180 word range, where the count reaches its highest at 1,065 essays." (p.15)
Q148	"The input to the model consisted of an instruction to score the essay along with the essay text, and the model was fine-tuned to generate the trait names followed by their corresponding scores." (p.16)
Q157	"The essay and prompt were provided as input to the encoder, with the max-pooled token representation concatenated with the handcrafted features." (p.16)
Q184	"The number of examples was fixed to 5, balancing scoring context with the context length limit (4096 tokens)." (p.18)
Q185	"When a prompt exceeded this limit, we iteratively removed examples until it fit." (p.18)

WEB-1	UAE teachers ~75% AI tool adoption per OECD TALIS 2024 (https://www.agbi.com/education/2025/10/uae-teachers-lead-world-in-classroom-ai-adoption/)
WEB-17	Saudi Arabia's Madrasati LMS deployed to all public K-12 schools, 92% student attendance (https://www.rowad-alkhaleej.edu.sa/en/online-learning-in-saudi-arabia/)

Information Gaps

- Whether handwritten-essay workflows remain dominant in non-Gulf target countries
- Length distribution of typical G10–12 essays in Egyptian, Jordanian, Lebanese, Moroccan curricula

Requires Expert Verification

- Country-by-country prevalence of digital vs. paper essay submission in G10–12 Arabic classrooms

Output Ontology (OO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: *LAILA* | ***Context:*** *Arab-Country G10–12 Arabic Teachers: AI Essay Grading Assistance*

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The output ontology consists of seven trait scores plus a holistic sum [Q34, Q35], with six traits on a 0–5 scale and REL on a 0–2 scale [Q36], grounded in the QUTC CAST rubric [Q33]. Three concerns drive the low score, two of which are flagged HIGH priority in elicitation. (1) The trait taxonomy is Qatari-curriculum-derived; no mapping is documented to Egyptian holistic grading [WEB-5], Tawjihi [WEB-10], Qiyas [WEB-7], CRDP, or Moroccan frameworks. Egypt's holistic-only practice means the analytic seven-trait label space may not be the construct teachers actually evaluate (elicitation Q2, Q4). (2) The output is exclusively discrete point-score labels [Q181, Q182]; the deployment requires confidence and rationale outputs [elicitation Q3], which are entirely absent from the output ontology. (3) The narrow REL 0–2 scale is acknowledged by the authors as limiting QWK granularity [Q91]. Strengths: the trait set is internally coherent, the REL=0 cascade rule [Q37] is explicit, and the rubric is documented in both Arabic [Q114] and English [Q116, Q118]. But for teacher-facing deployment across multi-country contexts, the output ontology is structurally misaligned on two of the highest-priority elicitation dimensions.

ID	Evidence
Q2	"The dataset comprises 7,859 essays across 8 distinct prompts making it the most extensive publicly available Arabic resource of its kind." (p.1)
Q3	"Under Institutional Review Board (IRB) approval, we visited 24 high schools in Qatar and collected essays directly from 4,372 students in authentic classroom settings." (p.1)
Q4	"Table 1: Brief description of the traits in LAILA." (p.1)
Q33	"All essays in LAILA dataset were scored using the Core Academic Skills Test (CAST) rubric [D7], developed by the Qatar University Testing Center (QUTC)." (p.5)
Q34	"The rubric covers 7 writing traits [D4]: Relevance (REL), Organization (ORG), Vocabulary (VOC), Style (STY), Development (DEV), Mechanics (MEC), and Grammar (GRA)." (p.5)
Q35	"Additionally, a Holistic score (HOL) was computed as the sum of the 7 trait scores." (p.5)
Q36	"Six traits (all but REL) were rated on a 6-point scale (0 = lowest, 5 = highest), while REL was rated on a 3-point scale (0 = not relevant, 1 = partially relevant, and 2 = fully relevant)." (p.5)
Q37	"Furthermore, if an essay received a REL score of 0, all other trait scores were set to 0, as irrelevant responses are not subject to further evaluation." (p.5)
Q91	"Fourth, the relevance trait is scored on a narrow scale from 0 to 2, which limits the granularity of evaluation and may cause QWK to underrepresent subtle differences in model predictions." (p.10)
Q114	"For annotating LAILA, we adopted the rubric from the Core Academic Skills Test (CAST), which is provided in Arabic, developed by the Qatar University Testing Center (QUTC)." (p.12)
Q116	"An English-translated version of the CAST rubric is presented in Table 6." (p.12)
Q118	"Table 6: CAST Persuasive/Argumentative Writing Rubric - English Translation (Bashendy et al., 2024)." (p.13)
Q181	"In the zero-shot setting, the LLM is tasked with generating trait scores in JSON format, given the prompt text, essay, and trait rubrics." (p.18)
Q182	"The holistic score was computed as the sum of the individual trait scores." (p.18)
WEB-5	Egypt's holistic grading culture suggests Egyptian teachers may apply different evaluative standards than Qatar-based annotators (https://grokipedia.com/page/academic_grading_in_egypt)

WEB-7	Saudi Qiyas focuses on higher-education entry testing rather than secondary multi-trait classroom rubrics (https://etec.gov.sa/en/centers/qiyas)
WEB-10	Jordan's Tawjihi Arabic exam emphasizes literary comprehension and grammar over trait-scored free composition (https://en.wikipedia.org/wiki/Tawjihi)

Strengths

- Seven-trait analytic schema aligns with rubric-based grading practice in Gulf countries (Qatar, UAE, Saudi Arabia) [Q34, WEB-7]
- Explicit rubric documentation (Arabic original + English translation) supports interpretability [Q114, Q116, Q118]
- Clear scoring scale and decision rules including REL=0 cascade [Q36, Q37]
- Score distributions reported per trait per prompt give deployment users a sense of label calibration [Q40, Q119]

ID	Evidence
Q34	"The rubric covers 7 writing traits [D4]: Relevance (REL), Organization (ORG), Vocabulary (VOC), Style (STY), Development (DEV), Mechanics (MEC), and Grammar (GRA)." (p.5)
Q36	"Six traits (all but REL) were rated on a 6-point scale (0 = lowest, 5 = highest), while REL was rated on a 3-point scale (0 = not relevant, 1 = partially relevant, and 2 = fully relevant)." (p.5)
Q37	"Furthermore, if an essay received a REL score of 0, all other trait scores were set to 0, as irrelevant responses are not subject to further evaluation." (p.5)
Q40	"Most traits follow a near-normal pattern on the 6-point scale, indicating overall positive essay quality. The REL scores show that 86% of the essays agree strongly with the prompt. In contrast, GRA and VOC show the highest percentage of score 1 at 16%, followed by MEC at 13%, highlighting that these traits pose the greatest challenges." (p.6)
Q114	"For annotating LAILA, we adopted the rubric from the Core Academic Skills Test (CAST), which is provided in Arabic, developed by the Qatar University Testing Center (QUTC)." (p.12)
Q116	"An English-translated version of the CAST rubric is presented in Table 6." (p.12)
Q118	"Table 6: CAST Persuasive/Argumentative Writing Rubric - English Translation (Bashendy et al., 2024)." (p.13)
Q119	"Figure 4 presents the score distributions in LAILA across 8 distinct prompts (P1–P8) and 7 writing traits, including Relevance (REL), Organization (ORG), Vocabulary (VOC), Style (STY), Development (DEV), Mechanics (MEC), and Grammar (GRA), alongside a Holistic (HOL) score representing the sum of all individual traits." (p.14)
WEB-7	Saudi Qiyas focuses on higher-education entry testing rather than secondary multi-trait classroom rubrics (https://etec.gov.sa/en/centers/qiyas)

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Categories are regionally relevant for analytic-rubric users (Gulf, UAE) but not for holistic-grading users (Egypt) [WEB-5]; cross-national rubric mapping is not documented [WEB-7, WEB-10].

OO-2: Missing categories for the deployment: (1) no holistic-only output mode for Egyptian-style grading; (2) no confidence/uncertainty category despite elicitation Q3 requiring it; (3) no rationale/justification category despite elicitation Q3 requiring it; (4) no narrative/descriptive trait specifications since those genres are excluded [Q89].

OO-3: The CAST rubric is QUTC-derived and may encode Gulf-curriculum stylistic and developmental norms in Style, Development, and Organization scoring; the elicitation flags this as warranting verification (Q1, OC priority) rather than dismissal. No web evidence documents how CAST descriptors compare with Maghrebi or Levantine rhetorical traditions.

OO-4: Stakeholder-driven taxonomy redesign is recommended but not undertaken in LAILA; the dataset's own ethics statement requires fairness auditing and stakeholder consultation before real-world application [Q106].

OO-5: Documented taxonomy issues harming structural and content validity: (1) discrete-label-only output form omitting confidence and rationale [Q181, Q182]; (2) narrow REL scale [Q91]; (3) no cross-national rubric mapping [WEB-5, WEB-7, WEB-10]; (4) no holistic-only mode [WEB-5].

ID	Evidence
Q1	“LAILA, the largest publicly available Arabic AES dataset to date, comprising 7,859 essays annotated with holistic and trait-specific scores on seven dimensions: relevance, organization, vocabulary, style, development, mechanics, and grammar.” (p.1)
Q3	“Under Institutional Review Board (IRB) approval, we visited 24 high schools in Qatar and collected essays directly from 4,372 students in authentic classroom settings.” (p.1)
Q33	“All essays in LAILA dataset were scored using the Core Academic Skills Test (CAST) rubric [D7], developed by the Qatar University Testing Center (QUTC).” (p.5)
Q34	“The rubric covers 7 writing traits [D4]: Relevance (REL), Organization (ORG), Vocabulary (VOC), Style (STY), Development (DEV), Mechanics (MEC), and Grammar (GRA).” (p.5)
Q89	“Second, LAILA includes only explanatory and persuasive writing prompts, limiting genre diversity and potentially affecting model robustness across other styles, such as narrative or descriptive writing.” (p.10)
Q91	“Fourth, the relevance trait is scored on a narrow scale from 0 to 2, which limits the granularity of evaluation and may cause QWK to underrepresent subtle differences in model predictions.” (p.10)
Q106	“We recommend that any system developed using LAILA undergo fairness auditing and stakeholder consultation before real-world application.” (p.10)
Q181	“In the zero-shot setting, the LLM is tasked with generating trait scores in JSON format, given the prompt text, essay, and trait rubrics.” (p.18)
Q182	“The holistic score was computed as the sum of the individual trait scores.” (p.18)
WEB-5	Egypt's holistic grading culture suggests Egyptian teachers may apply different evaluative standards than Qatar-based annotators (https://grokipedia.com/page/academic_grading_in_egypt)
WEB-7	Saudi Qiyas focuses on higher-education entry testing rather than secondary multi-trait classroom rubrics (https://etec.gov.sa/en/centers/qiyas)
WEB-10	Jordan's Tawjihi Arabic exam emphasizes literary comprehension and grammar over trait-scored free composition (https://en.wikipedia.org/wiki/Tawjihi)
WEB-13	RaDME generates trait scores with co-produced rationales using knowledge distillation — English only, no Arabic adaptation documented (https://arxiv.org/html/2502.20748v1)

Information Gaps

- Whether CAST trait descriptors carry Gulf-specific stylistic norms detectable in score distributions
- Whether any Arab country's national rubric maps onto a subset of the seven traits

Requires Expert Verification

- Whether Egyptian, Lebanese, or Moroccan teachers would re-weight or re-define Style and Development categories given their professional norms

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Remediation

Gap 1: No category for confidence/uncertainty or rationale outputs, despite both being HIGH-priority deployment requirements [elicitation Q3]

Recommendation: Extend the output ontology with two new categories: (1) per-trait confidence/uncertainty signal, and (2) rationale text. Adapt an English rationale-generation framework (RaDME [WEB-13] or RMTS [WEB-12]) to Arabic with teacher-facing evaluation.

ID	Evidence
Q3	“Under Institutional Review Board (IRB) approval, we visited 24 high schools in Qatar and collected essays directly from 4,372 students in authentic classroom settings.” (p.1)
WEB-12	BERT-based Arabic AES system documented but without rationale generation (https://www.researchgate.net/publication/382302599_Automated_essay_scoring_in_Arabic_a_dataset_and_analysis_of_a_BERT-based_system)
WEB-13	RaDME generates trait scores with co-produced rationales using knowledge distillation — English only, no Arabic adaptation documented (https://arxiv.org/html/2502.20748v1)

Output Content (OC)

Score: 3/5 — Moderate gaps | Confidence: medium

***Benchmark:** LAILA* | ***Context:** Arab-Country G10–12 Arabic Teachers: AI Essay Grading Assistance*

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Annotation procedures are well-documented and rigorous: 6 annotators and 3 supervisors, all Arabic language teachers/lecturers with five holding MSc/PhD in Arabic [Q30, Q31]; structured training [Q102]; comprehensive guidebook with exemplars [Q41]; annotator blinding to student identity [Q42]; double scoring with mean-rounded-down aggregation and supervisor escalation at HOL ≥ 6 [Q43]; QWK IAA of 0.66–0.75 classified as substantial [Q44]; fair compensation [Q101]. These are real strengths. The validity concern is that all annotators appear to be Qatar-based and trained on QUTC CAST [Q30, Q33], with no documentation of nationality diversity (gap_id 5: no documentation found). The elicitation marks OC as MODERATE because the user judges pan-Arab MSA standards uniform for surface linguistic quality (Q1), which lowers — but does not eliminate — the risk that evaluative judgments on Style, Development, and Organization carry implicit Gulf-curriculum norms. The aggregation method (mean-rounded-down) [Q43] could erase minority interpretations on borderline cases. P5 had a 23.3% third-annotator escalation rate [Q45], suggesting some prompts produced systematic annotator disagreement. Overall, methodologically strong but with an unresolved annotator-pool diversity question that matters for cross-country deployment.

ID	Evidence
Q1	“LAILA, the largest publicly available Arabic AES dataset to date, comprising 7,859 essays annotated with holistic and trait-specific scores on seven dimensions: relevance, organization, vocabulary, style, development, mechanics, and grammar.” (p.1)
Q30	“The hired annotation team consisted of 6 annotators and 3 supervisors, all of whom were Arabic language teachers or lecturers with educational backgrounds.” (p.5)
Q31	“Five members of the team hold advanced degrees (MSc or PhD) in the Arabic language.” (p.5)
Q33	“All essays in LAILA dataset were scored using the Core Academic Skills Test (CAST) rubric [D7], developed by the Qatar University Testing Center (QUTC).” (p.5)
Q41	“To ensure consistency, 2 supervisors developed an annotation guidebook with detailed terminology, exemplars, and annotated practices for each prompt type. Annotators were required to review these materials and complete training sessions before formal annotation. Moderation sessions followed, where discrepancies were discussed and interpretations of the rubric were harmonized.” (p.6)
Q42	“The scoring process was conducted using the Assessment Gourmet Platform, which supports large-scale, anonymized essay scoring. The platform ensured that annotators were blinded to student identity [D6], while only supervisors had access to annotator metadata for monitoring purposes.” (p.6)
Q43	“The essays were randomly distributed among the annotators to minimize systematic bias, and the scoring sessions were capped to prevent annotators' fatigue. Each essay was independently scored by two annotators across all traits. If the difference in HOL scores between the two annotators was less than 6 points, the mean of the two scores was computed and then rounded down to the nearest integer; this rounded value was adopted as the final score for each trait. However, essays with large discrepancies between annotators (≥ 6 points difference in the HOL score) were flagged and escalated to a supervisor (a third annotator in this case), who provided the final adjudicated score and offered feedback to the annotators to improve alignment and consistency in subsequent batches of essays.” (p.6)
Q44	“We compute IAA using Quadratic Weighted Kappa (QWK) (Williamson et al., 2012) to assess agreement between the two main annotators (A1 and A2), prior to adjudication. Table 4 reports QWK per trait and prompt, with agreement strength following Landis and Koch (1977). Overall, IAA was substantial across prompts, with an average QWK ranging from 0.66 (P5) to 0.75 (P1), showing strong consistency.” (p.6)
Q45	“However, variability emerges by prompt and trait. In particular, P5 showed the lowest average agreement, noting that it has the highest percentage of essays that require a third annotator (A3: 23.3%; Table 3). This suggests greater initial disagreement” (p.6)

Q101	“All annotators were qualified Arabic language educators who received fair compensation at or above local professional rates.” (p.10)
Q102	“They completed structured training on annotation guidelines, and inter-annotator disagreements were resolved through systematic adjudication procedures.” (p.10)

Strengths

- Annotators are qualified Arabic-language educators, five with advanced degrees [Q30, Q31]
- Structured training, annotation guidebook, and moderation sessions [Q41, Q102]
- Double annotation with explicit adjudication procedure for disagreements ≥ 6 HOL points [Q43]
- Substantial inter-annotator agreement (QWK 0.66–0.75) [Q44]
- Annotator blinding to student identity reduces demographic bias [Q42, Q98]
- Fair compensation at or above local professional rates [Q101]
- IRB approval and informed consent documented [Q9, Q95]

ID	Evidence
Q9	“IRB Number: QU-IRB 159/2024-EA” (p.1)
Q30	“The hired annotation team consisted of 6 annotators and 3 supervisors, all of whom were Arabic language teachers or lecturers with educational backgrounds.” (p.5)
Q31	“Five members of the team hold advanced degrees (MSc or PhD) in the Arabic language.” (p.5)
Q41	“To ensure consistency, 2 supervisors developed an annotation guidebook with detailed terminology, exemplars, and annotated practices for each prompt type. Annotators were required to review these materials and complete training sessions before formal annotation. Moderation sessions followed, where discrepancies were discussed and interpretations of the rubric were harmonized.” (p.6)
Q42	“The scoring process was conducted using the Assessment Gourmet Platform, which supports large-scale, anonymized essay scoring. The platform ensured that annotators were blinded to student identity [D6], while only supervisors had access to annotator metadata for monitoring purposes.” (p.6)
Q43	“The essays were randomly distributed among the annotators to minimize systematic bias, and the scoring sessions were capped to prevent annotators' fatigue. Each essay was independently scored by two annotators across all traits. If the difference in HOL scores between the two annotators was less than 6 points, the mean of the two scores was computed and then rounded down to the nearest integer; this rounded value was adopted as the final score for each trait. However, essays with large discrepancies between annotators (≥ 6 points difference in the HOL score) were flagged and escalated to a supervisor (a third annotator in this case), who provided the final adjudicated score and offered feedback to the annotators to improve alignment and consistency in subsequent batches of essays.” (p.6)
Q44	“We compute IAA using Quadratic Weighted Kappa (QWK) (Williamson et al., 2012) to assess agreement between the two main annotators (A1 and A2), prior to adjudication. Table 4 reports QWK per trait and prompt, with agreement strength following Landis and Koch (1977). Overall, IAA was substantial across prompts, with an average QWK ranging from 0.66 (P5) to 0.75 (P1), showing strong consistency.” (p.6)
Q95	“The collection of LAILA received approval from an Institutional Review Board (IRB Number: QU-IRB 159/2024-EA) and was conducted with informed consent from both students and their legal guardians.” (p.10)
Q98	“Annotation was conducted on a platform that prevented annotators from accessing student identities or demographic information.” (p.10)
Q101	“All annotators were qualified Arabic language educators who received fair compensation at or above local professional rates.” (p.10)
Q102	“They completed structured training on annotation guidelines, and inter-annotator disagreements were resolved through systematic adjudication procedures.” (p.10)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement

OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Partially. Ground-truth labels reflect qualified Arabic-educator judgment [Q30] but specifically Qatar-based educators trained on QUTC CAST [Q33]; whether they reflect Egyptian, Levantine, or Maghrebi educator perspectives is undocumented.

OC-2: Potential disagreement is plausible on Style, Development, and Organization given different national rhetorical traditions; the elicitation flags this as warranting verification (Q1). The pan-Arab MSA assumption should hold for surface mechanics/grammar but is less certain for evaluative traits.

OC-3: Annotator demographics are partially documented — qualifications and degree level [Q30, Q31] are reported, but nationality, dialect background, and curricular training origin beyond CAST are not. No Datasheet or Data Statement section specifies these.

OC-4: INSUFFICIENT DOCUMENTATION — no re-annotation by representative non-Qatari Arab annotators is reported; the LAILA paper does not include any cross-country annotator validation study.

OC-5: Aggregation is mean-of-two-scores rounded down to nearest integer [Q43], with supervisor adjudication only when HOL discrepancy ≥ 6 points. This may erase legitimate minority perspectives on borderline cases (e.g., a 3 vs. 4 disagreement is averaged and rounded down to 3 without further review).

OC-6: Documented label-correctness considerations: (1) annotator pool nationality undocumented [gap_id 5]; (2) substantial but not perfect IAA, with P5 showing 23.3% escalation [Q45]; (3) round-down aggregation may bias scores slightly downward on disagreement cases [Q43].

ID	Evidence
Q1	"LAILA, the largest publicly available Arabic AES dataset to date, comprising 7,859 essays annotated with holistic and trait-specific scores on seven dimensions: relevance, organization, vocabulary, style, development, mechanics, and grammar." (p.1)
Q30	"The hired annotation team consisted of 6 annotators and 3 supervisors, all of whom were Arabic language teachers or lecturers with educational backgrounds." (p.5)
Q31	"Five members of the team hold advanced degrees (MSc or PhD) in the Arabic language." (p.5)
Q33	"All essays in LAILA dataset were scored using the Core Academic Skills Test (CAST) rubric [D7], developed by the Qatar University Testing Center (QUTC)." (p.5)
Q43	"The essays were randomly distributed among the annotators to minimize systematic bias, and the scoring sessions were capped to prevent annotators' fatigue. Each essay was independently scored by two annotators across all traits. If the difference in HOL scores between the two annotators was less than 6 points, the mean of the two scores was computed and then rounded down to the nearest integer; this rounded value was adopted as the final score for each trait. However, essays with large discrepancies between annotators (≥ 6 points difference in the HOL score) were flagged and escalated to a supervisor (a third annotator in this case), who provided the final adjudicated score and offered feedback to the annotators to improve alignment and consistency in subsequent batches of essays." (p.6)
Q45	"However, variability emerges by prompt and trait. In particular, P5 showed the lowest average agreement, noting that it has the highest percentage of essays that require a third annotator (A3: 23.3%; Table 3). This suggests greater initial disagreement" (p.6)
WEB-5	Egypt's holistic grading culture suggests Egyptian teachers may apply different evaluative standards than Qatar-based annotators (https://grokipedia.com/page/academic_grading_in_egypt)

Information Gaps

- Annotator nationality and dialectal background distribution (gap_id 5: no documentation found)
- Whether the round-down aggregation introduces systematic downward bias on borderline scores
- What proportion of supervisor adjudications involved trait-specific (vs. holistic) disagreements

Requires Expert Verification

- Whether non-Qatari Arab Arabic-language educators would assign systematically different Style, Development, or Organization scores on the same essays
- Inquiry to LAILA authors regarding annotator nationality composition

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Remediation

Gap 1: Annotator nationality diversity within the Qatar-based pool is undocumented [gap_id 5]; cross-country annotator agreement is unknown

Recommendation: Direct inquiry to LAILA authors for annotator nationality composition; commission a small cross-country re-annotation study (~500 essays) with Egyptian, Saudi, Lebanese, and Moroccan Arabic-language educators to quantify QWK drift on Style, Development, and Organization.

Output Form (OF)

Score: 1/5 — Serious concern | Confidence: high

***Benchmark:** LAILA* | ***Context:** Arab-Country G10–12 Arabic Teachers: AI Essay Grading Assistance*

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

This is the dimension with the most direct deployment mismatch and is flagged HIGH priority in elicitation. LAILA's evaluation produces only discrete point-score labels [Q181, Q182] evaluated by Quadratic Weighted Kappa [Q58]. The benchmark does not evaluate confidence calibration, uncertainty quantification, score-range prediction, or rationale generation. The deployment use case explicitly requires confidence signals and explanatory justification [elicitation Q3] so teachers can intervene meaningfully on borderline essays. There is no overlap between the benchmark's evaluated output form and the deployment's required output form for these features. Web research confirms this is a structural gap in the field: rationale-generation frameworks (RMTS, RaDME, QwenScore+) exist for English but no Arabic adaptation has been documented [WEB-12, WEB-13, WEB-14], and TAQEEM 2025 used Chain-of-Thought prompting but focused on accuracy, not rationale [WEB-15]. The benchmark's own ethics statement explicitly prohibits use 'for high-stakes educational decisions affecting individual students' [Q104] and warns against 'deployment in operational assessment contexts without independent validation' [Q105]. Combined with the elicitation marking both OO and OF as HIGH priority for this exact reason, the score is 1 — fundamental misalignment.

ID	Evidence
Q3	“Under Institutional Review Board (IRB) approval, we visited 24 high schools in Qatar and collected essays directly from 4,372 students in authentic classroom settings.” (p.1)
Q58	“We evaluate model performance using QWK to measure agreement between human and model scores.” (p.8)
Q104	“We explicitly prohibit its use for high-stakes educational decisions affecting individual students.” (p.10)
Q105	“Users must commit to: (1) preventing re-identification attempts, (2) avoiding demographic profiling or inference, and (3) refraining from deployment in operational assessment contexts without independent validation.” (p.10)
Q181	“In the zero-shot setting, the LLM is tasked with generating trait scores in JSON format, given the prompt text, essay, and trait rubrics.” (p.18)
Q182	“The holistic score was computed as the sum of the individual trait scores.” (p.18)
WEB-12	BERT-based Arabic AES system documented but without rationale generation (https://www.researchgate.net/publication/382302599_Automated_essay_scoring_in_Arabic_a_dataset_and_analysis_of_a_BERT-based_system)
WEB-13	RaDME generates trait scores with co-produced rationales using knowledge distillation — English only, no Arabic adaptation documented (https://arxiv.org/html/2502.20748v1)
WEB-14	QwenScore+ uses rubric-aware Chain-of-Thought prompting with RLHF — English only (https://www.preprints.org/manuscript/202504.2338)
WEB-15	TAQEEM 2025 used CoT prompting on Arabic essays but focused on scoring accuracy, not rationale output (https://aclanthology.org/2025.arabicnlp-sharedtasks.135.pdf)

Strengths

- QWK as the evaluation metric matches the metric used for IAA, providing internal coherence between human and model agreement [Q44, Q58]
- Three-seed repetition for LLM experiments supports reporting variability [Q189]
- Structured JSON output enforced via outlines library [Q190] could be extended to include rationale fields

ID	Evidence

Q44	"We compute IAA using Quadratic Weighted Kappa (QWK) (Williamson et al., 2012) to assess agreement between the two main annotators (A1 and A2), prior to adjudication. Table 4 reports QWK per trait and prompt, with agreement strength following Landis and Koch (1977). Overall, IAA was substantial across prompts, with an average QWK ranging from 0.66 (P5) to 0.75 (P1), showing strong consistency." (p.6)
Q58	"We evaluate model performance using QWK to measure agreement between human and model scores." (p.8)
Q189	"To account for variability in example selection, each experiment was repeated three times with random seeds 1, 11, and 42, and we report the average performance across runs." (p.18)
Q190	"For all the LLM experiments, we used vLLM for inference with the outlines library to enforce the JSON output format." (p.18)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: No. Benchmark output is discrete trait scores plus holistic sum [Q181, Q182]; deployment requires per-trait scores plus confidence signal plus rationale text. The benchmark does not evaluate the latter two.

OF-2: Not applicable — neither benchmark nor deployment requires speech output. Output is text only on both sides.

OF-3: INSUFFICIENT DOCUMENTATION on this specific point in the LAILA paper, but the deployment population is qualified teachers; literacy/accessibility considerations apply more to student-facing outputs than to the teacher-facing scoring interface evaluated here.

OF-4: Documented form mismatches harming external validity: (1) no confidence/uncertainty output evaluated [Q181]; (2) no rationale output evaluated [Q181, Q182]; (3) no in-domain Arabic precedent for rationale-generation evaluation [WEB-12, WEB-13, WEB-15]; (4) benchmark explicitly prohibits high-stakes operational use without independent validation [Q104, Q105].

ID	Evidence
Q104	"We explicitly prohibit its use for high-stakes educational decisions affecting individual students." (p.10)
Q105	"Users must commit to: (1) preventing re-identification attempts, (2) avoiding demographic profiling or inference, and (3) refraining from deployment in operational assessment contexts without independent validation." (p.10)
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Information Gaps

- Whether any Arabic-language AES system has been validated with teacher-facing rationale output (gap_id 3: not found)
- Whether Arabic teachers would accept LLM-generated rationales as professionally credible

Requires Expert Verification

- Acceptance of AI-generated trait scores and rationales by Arab-country secondary Arabic teachers across the deployment range

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Remediation

Gap 1: Benchmark evaluates only discrete point-score outputs via QWK; deployment requires uncertainty signals and rationale text [Q58, Q181]

Recommendation: Define and report new metrics alongside QWK: (1) calibration metrics for confidence outputs (e.g., expected calibration error against teacher disagreement); (2) rationale quality metrics evaluated by qualified Arab teachers across multiple countries. Treat current QWK results as necessary but not sufficient evidence.

ID	Evidence
Q58	“We evaluate model performance using QWK to measure agreement between human and model scores.” (p.8)
Q181	“In the zero-shot setting, the LLM is tasked with generating trait scores in JSON format, given the prompt text, essay, and trait rubrics.” (p.18)